Example 9.49

Find the area of the region bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Solution

The ellipse is symmetric about both major and minor axes. It is sketched as in Fig.9.16. So, viewing in the positive direction of y-axis, the required area A is four times the area of the region bounded by the portion of the ellipse in the first quadrant $\left(y=\frac{b}{a}\sqrt{a^2-x^2}, 0<x<a\right)$, x-axis, $x=0$ and $x=a$.

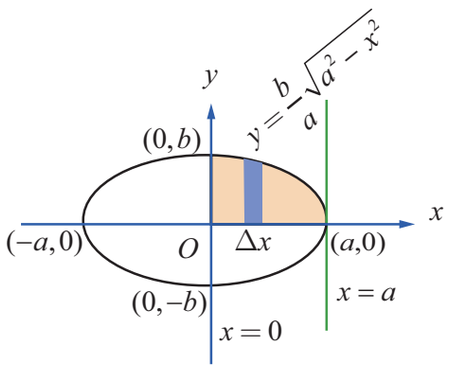


Fig. 9.16

Hence, by taking vertical strips, we get

$$A = 4\int_0^a y\,dx = 4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}\,dx$$

$$= \frac{4b}{a}\left[\frac{x\sqrt{a^2-x^2}}{2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a = \frac{4b}{a}\times\frac{\pi a^2}{4} = \pi ab$$

Note

Viewing in the positive direction of x-axis, the required area A is four times the area of the region bounded by the portion of the ellipse in the first quadrant $\left(x=\frac{a}{b}\sqrt{b^2-y^2}, 0<y<b\right)$ y-axis, $y=0$ and $y=b$. Hence, by taking horizontal strips (see Fig.9.17), we get

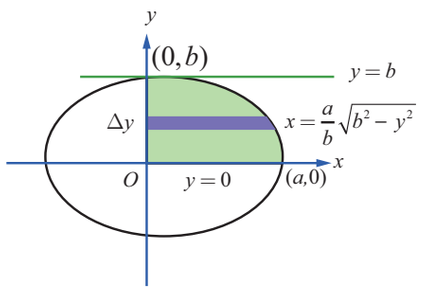


Fig. 9.17

$$A = \int_0^a x\,dy = 4\int_0^b \frac{a}{b}\sqrt{b^2-y^2}\,dy$$

$$= \frac{4a}{b}\left[\frac{y\sqrt{b^2-y^2}}{2}+\frac{b^2}{2}\sin^{-1}\left(\frac{y}{b}\right)\right]_0^b = \frac{4a}{b}\times\frac{\pi b^2}{4} = \pi ab.$$

Note

Putting $b=a$ in the above result, we get that the area of the region enclosed by the circle $x^2+y^2=a^2$ is πa^2.

Example 9.50

Find the area of the region bounded between the parabola $y^2=4ax$ and its latus rectum.

Solution

The equation of the latus-rectum is $x=a$. It intersects the parabola at the points $L(a,2a)$ and $L_1(a,-2a)$. The required area is sketched in Fig. 9.18. By symmetry, the required area A is twice the area bounded by the portion of the parabola

$$y=2\sqrt{a}\sqrt{x}, x\text{-axis}, x=0 \text{ and } x=a.$$

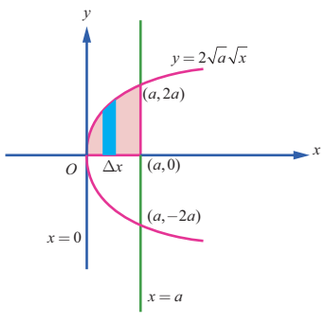


Fig. 9.18

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